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Advanced Reactor Materials Test 1

1) From inside to out, the five layers are:

- Fuel Kernel: Responsible for doing the fission and producing heat. Has uranium carbides/oxides.
- Buffer layer: Porous graphite layer that provides a mechanical buffer between the fuel kernel and IP₁C layer. Also retains fission products.
- Inner Pyrolytic Carbon (IP₁C): Protects SiC layer from fission product attack, puts SiC layer in compression.
- SiC layer: Main pressure boundary and fission product barrier. Silicon Carbide. Hardest layer.
- Outer Pyrolytic Carbon (OP₁C): Also puts SiC layer in compression. ~ protects SiC during fab.

2) Oxides are more easily produced (oxidation not a concern) and more chemically stable. Carbides are more compatible with carbon layers at high T. UO_2 use leads to CO production, but UC_2 use leads to poor fission product retention since FPs can't form oxides. For a balance, an 80/20 UO_2/UC_2 split is used.

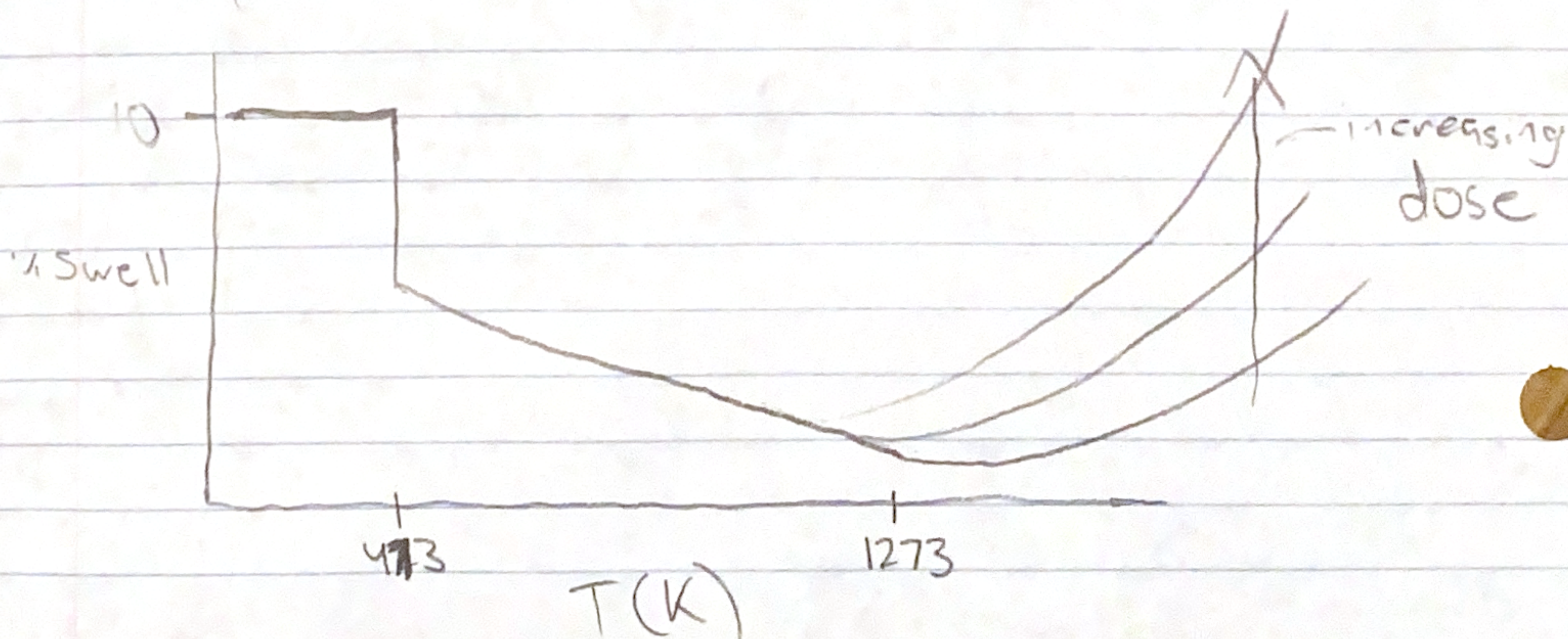
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3) Fission products can be a variety of elements, forming insoluble oxides, soluble oxides, metallic precipitates, inert gases, and volatiles. Insoluble products may diffuse out of the kernel. Ag and Pd are of particular concern. Pd chemically attacks the S.C. layer while Ag may diffuse through it. - needed more

4) There are 3 regions:

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• Low T ($< 473\text{K}$): amorphization can occur from irradiation, causing swelling as high as $\sim 10\%$.

• Med T ($473\text{K} - 1273\text{K}$): Point defect region. Swelling occurs from point defects, with higher irradiation temperature resulting in less swelling. Swelling increases logarithmically with dose before reaching saturation. Irradiation damage is reduced at high T irradiation due to recombination of defects.

• High T ($> 1273\text{K}$): Void swelling region. Voids are now capable of forming, which dominates swelling. Swelling increases with irradiation temperature and dose.



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5) In the point defect saturation region, thermal resistivity from irradiation is directly linear with swelling. This is because point defects and grain boundaries are phonon scatterers. Irradiation damage causes point defects, which both scatter phonons and cause swelling, hence the relationship. Thermal conductivity is inversely related to this resistivity.

In the high T region, the linear relationship does not exist. Voids do not scatter phonons as strongly as point defects.

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6) Graphite first shrinks and then swells. During manufacturing, anisotropic CTEs in graphite lead to the formation of Mozowski cracks, which run along basal (a) planes.

Irradiation damage often creates an interstitial between basal planes, causing an expansion perpendicular to the planes (c direction) and a contraction in the a direction.

At first, the ^{Mozowski} cracks allow room for the c direction expansion, leading to net shrinkage. Once the cracks are "filled up", the expansion outweighs the a direction contraction, resulting in net swelling.

"Turnaround" (when shrink $\rightarrow$ swell) occurs sooner at higher irradiation temperature.

inside the particle

7) • Overpressure: Pressure from gas buildup[^] causes tensile stress in the SiC that is overwhelming, leading to failure.

• IPYC cracking: IPYC fails due to carbon shrinkage, negating its ability to compress the SiC to relieve SiC tension, leading to a crack in the SiC

• Fission product attack: Fission products diffuse through the IPYC and chemically attack the SiC to the point of failure. Pd is commonly the culprit.

8) Use of ZrC instead of SiC as the main pressure boundary. ZrC is more stable at very high temperatures and more resistant to FP attack, but it is not as resistant to oxidation.

9) Main concerns are oxidation, carburization, decarburization, and breakdown of the passive oxide layer at high T. Nickel based alloys are used for corrosion resistance. There must be enough oxygen and carbon in the helium stream to keep the surface of the alloy stable (Zone III). Materials are only fully qualified up to 750°C, which is a problem for VHTRs. - almost all